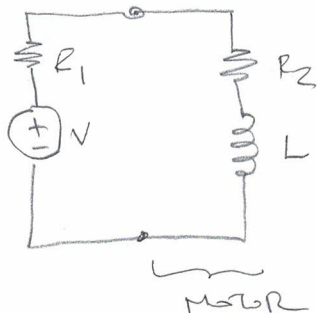


7.21)



$$V = 120 \angle 0^\circ \quad \omega = 377 \text{ rad/sec}$$

$$R_1 = 2 \Omega$$

$$R_2 = 10 \Omega$$

$$L = 20 \cdot 10^{-3} \text{ H}$$

a) P_f ? BEST FROM SPANNING SHOWN.

$$P_f = \frac{P}{V_{\text{rms}} I_{\text{rms}}} = \cos(\phi_v - \phi_i)$$

$$\begin{aligned} \text{Dir } I_{\text{rms}} &= \frac{V_{\text{rms}}}{R_1 + R_2 + j\omega L} = \frac{120 \angle 0^\circ}{12 + j377 \cdot 20 \cdot 10^{-3}} = \\ &= \frac{120 \angle 0^\circ}{14.17 \angle 32.14^\circ} = \underline{8.47 \angle -32.14^\circ \text{ A rms}} \end{aligned}$$

$$P_f = \cos(\phi_v - \phi_i) = \cos(0^\circ + 32.14^\circ) = \underline{0.847}$$

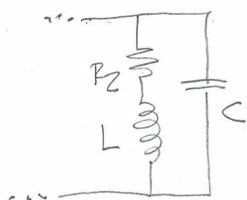
b) FOR MOTOR WHEN AT PASSIVE INK ANOAS
AS R_1 IN EN RESISTANCE $\Rightarrow P_{\text{motor}} = P_{\text{india}}$

c) MOTOR EFFICIENCY CONSUMPTION?

$$\begin{aligned} S_{\text{motor}} &= V_{\text{rms}} I_{\text{rms}}^* = Z_{\text{motor}} I_{\text{rms}} I_{\text{rms}}^* = \\ &= |I_{\text{rms}}|^2 Z_{\text{motor}} = 8.47^2 (10 + j377 \cdot 20 \cdot 10^{-3}) = \\ &= 717.41 + j540.93 \end{aligned}$$

$$\Rightarrow \underline{P = 717.41 \text{ W}}$$

d)



HUR STOR SKALL C VART FÖR
ATT ANOAS P_f TIL 0.9?

IDER TIL FÖRER BLIR



GEOMETRISKT

$$\phi_2 = \arctan\left(\frac{Q_2}{P}\right)$$

$$\phi_1 = 32.14^\circ \Rightarrow Q_1 = P \tan(\phi_1) = 717.41 \cdot \tan(32.14^\circ) =$$
$$= \underline{450.72 \text{ VAR}}$$

OCH

$$P_{f2} = \cos(\phi_2) \Rightarrow \phi_2 = \arccos(P_{f2}) = \cos^{-1}(0.9) =$$
$$= \underline{25.84^\circ}$$

$$\Rightarrow Q_2 = P \tan(\phi_2) = 717.41 \tan(25.84^\circ) =$$
$$= \underline{347.43 \text{ VAR}}$$

OCH Q2 MOTIVARAR

$$\Delta Q = Q_1 - Q_2 = V_{\text{RMS}}^2 \omega C \Rightarrow C = \frac{\Delta Q}{V_{\text{RMS}}^2 \omega}$$

MEN DET ÄR EN SPÄNNINGEN ÖVER KOND SOM SKALL ANVÄNDAS

VI VET ATT

$$V_{\text{RMS}} = V - R I_{\text{RMS}} = 120 \angle 0^\circ - 2.847 \angle -32.14^\circ =$$
$$= 105.65 - j9.01 \text{ V}_{\text{RMS}}$$
$$= 106.03 \angle -4.87^\circ$$

AUTIS

$$C = \frac{\Delta Q}{V_{\text{RMS}}^2 \omega} = \frac{(450.72 - 347.43)}{106.03^2 \cdot 377} = \underline{25.31 \cdot 10^{-6} \text{ F}}$$

$$C = \underline{25.3 \text{ nF}}$$

SVAR:

a) $P_f = 0.847$

b) $P_f = 0.847$

c) $P = 717.41 \text{ W}$

d) $C = 25.3 \text{ nF}$